Aanchala Ravindra Bhongade

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Professional Summary

MSc Data Science graduate with strong quantitative and Python skills, experienced in experimental design, multi-seed model evaluation under distribution shift, and large-scale data processing. Proven ability to design controlled experiments, diagnose failure modes, and extract reliable signals from complex data. Keen to apply rigorous statistical learning, feature identification, and careful data analysis to systematic trading and quantitative research.

Education

University of Bristol Sep 2025 – Sep 2026
MSc Data Science *Bristol, United Kingdom*

- Relevant study: Statistical Computing and Empirical Methods; Programming and Algorithms; AI and Text Analytics; Large-Scale Data Engineering; Data Science Methods and Practice; Visual Analytics.

Veermata Jijabai Technological Institute (VJTI) Dec 2021 – Jun 2025
B.Tech. Production Engineering, First Class, CGPA: 7.33/10 *Mumbai, India*

- Relevant study: Probability and Statistics; Operations Research; Business Analytics; Programming and Problem Solving in C++; Supply Chain Management.

Technical Skills

Quantitative Methods & Evaluation: Statistics, probability, experimental design, multi-seed validation, model evaluation under distribution shift, uncertainty quantification, feature identification, time-series analysis

Machine Learning & Analysis: Python (pandas, NumPy, scikit-learn, PyTorch, statsmodels), statistical testing, anomaly detection, Bayesian methods

Data Processing: Large-scale data cleaning, transformation, normalisation, preprocessing pipelines

Systems & Tools: AWS (EC2, S3, SQS, DynamoDB, CloudWatch, Auto Scaling), SQL, Git, Linux, Jupyter

Visualisation: Tableau (interactive dashboards, geographic mapping)

Relevant Experience

University of Bristol June 2026 – August 2026
MSc Research Dissertation – Length Generalisation in Transformer Sequence Models *Bristol, United Kingdom*

- Designed and executed 45 controlled multi-seed experiments on Transformer sequence models (3 algorithmic tasks, 5 positional encodings) using PyTorch and AWS GPU infrastructure to evaluate generalisation under increased sequence lengths (up to 1,024) with statistical validation across seeds.
- Built an evidence-based failure taxonomy distinguishing training underfitting, seed instability, immediate extrapolation collapse, and position-dependent error patterns, supported by position-wise error profiles and attention statistics.
- Established a multi-seed baseline-eligibility framework so only stable configurations were used for further comparison; separated interventions that preserved training performance from those that produced no observed improvement on unseen lengths.
- Evaluated mixed-length training, curriculum learning, and randomised padding; characterised task-specific failure modes using carry-condition analysis and dependency-distance diagnostics.
- Produced clear documentation of methods, failure modes, and limitations to support interpretation beyond aggregate accuracy.

University of Cambridge Sep 2024 – Mar 2025
Visiting Researcher – Cambridge Collective Intelligence and Design Group *Cambridge, United Kingdom*

- Built a Python pipeline integrating World Bank CMIP6 climate and demographic projections across five SSP scenarios with appliance-sales data to analyse heat exposure and cooling-energy demand in India through 2100.
- Evaluated two genders and 20 five-year age groups across four future periods, generating 40 scenario-specific contour plots alongside heat-index trends, percentile bounds, and Cooling Degree Day correlation analyses.
- Identified a rise in heat index to approximately 89°F by 2099 under SSP5-8.5, with higher cooling demand among older age groups and substantially wider uncertainty under high-emission scenarios; translated the work into a single-author peer-reviewed publication.

University of Zurich, *Relic-Kappa Project* Jul 2024 – Jan 2025
Undergraduate Scientific Computing Researcher *Zurich, Switzerland (Remote)*

- Processed FITS/HEALPix weak-lensing convergence maps on the UZH Science Cluster across six simulations, five redshifts ($z = 0.6$–2.0), and four Newtonian/relativistic configurations, covering 120 configuration-redshift combinations.
- Developed Python workflows using Astropy, Healpy/Anafast, NumPy, and Matplotlib to compute angular power spectra and produce analysis-ready map statistics and visualisations for downstream research.
- Compared Newtonian and relativistic outputs using mean and relative-difference calculations, 20-bin aggregation, standard deviations, and error bars across simulation runs.

Veermata Jijabai Technological Institute Oct 2023 – May 2024
Research Intern *Mumbai, India*

- Collaborated to identify the optimal EV charger using Multiple Criteria Decision Making (MCDM), applying MCDM frame-works and Super Decision Software to structure complex multi-variable evaluation, involving industry expert consultation,manufacturing plant visits, and extensive literature review.

Technical Projects

Socio-Economic Trends Study – Heating Deprivation & Health Outcomes (England & Wales)
- *Tableau, Python (pandas, scikit-learn), Bayesian methods*
- Harmonised 2011 and 2021 census data across local authorities and imputed missing values using Bayesian Ridge regression.
- Identified private rental tenure as the strongest predictor of heating deprivation (standardised coefficient 1.43, 95% CI [1.34, 1.52]) and higher no-heat rates among the 65+ age group.
- Built interactive Tableau dashboards and geographic maps that revealed spatial patterns in heating access and their relationship with health outcomes, enabling clear comparison between 2011 and 2021.

Elastic Event-Driven Data Pipeline on AWS
- *EC2, S3, SQS, DynamoDB, IAM, CloudWatch, Auto Scaling*
- Built a scalable event-driven processing system using S3, SQS, EC2 workers and DynamoDB with least-privilege IAM roles and CloudWatch monitoring.
- Implemented queue-depth-based Auto Scaling; load-tested with a 120-file burst that scaled workers from one to two, processed the full workload without data loss, and returned to minimum capacity, demonstrating reliable production controls.

Robust Prediction Intervals Under Distribution Shift
- *Research Proposal*
- Designed a controlled quantitative evaluation of four probabilistic forecasters with native and conformal calibration across four temporal shift types and three severity levels, assessing coverage, interval width, CRPS, and Winkler score.

Blockchain Security Analysis
- *Python, Pandas, Seaborn, SciPy, APIs (DeFiLlama, Binance)*
- Built a data pipeline using DeFiLlama and Binance APIs to merge Ethereum market metrics with historical hack data, executing statistical tests to quantify how market cycles influence crypto-crime severity.

Publication

Bhongade, A. (2025). *The Unequal Impact of Lethal Heatwaves in India for Males and Females: Insights from Climate Projection Data. Cambridge Journal of Climate Research*, 2(1), 182–194. DOI: 10.60866/CAM.241

- Sole-authored a peer-reviewed study integrating CMIP6 climate and demographic projections across five SSP scenarios to analyse heat exposure, cooling-energy demand, uncertainty, and age- and gender-based disparities in India through 2100.

Extracurricular

- AWS Academy Cloud Foundations; HarvardX R Basics
- CBSE Certificate of Merit for ranking in the top 0.1% nationally in Social Science.
- **Department Manager, Sports Fest VJTI:** led volunteer logistics for 500+ participants. | **Public Relations Executive, Tech Fest VJTI:** supported engagement for a national technical festival with 60,000+ attendees.
- **University Basketball Team Captain:** captained the team across seven matches, securing three inter-college wins.